

MULTIVARIABLE CALCULUS & HISTORY OF MODERN INDIA

Examination Year: 2026 | Maximum Marks: 75 | Total Questions: 3 | Time Allowed: 3 Hours

GENERAL INSTRUCTIONS:

1. All questions are compulsory unless specified otherwise.
2. Candidates must show all working, intermediate steps, and derivations to earn full marks.
3. Draw diagrams wherever necessary; label all parts clearly.
4. Figures, graphs and diagrams are provided with the respective question.
5. Use of scientific calculator is permitted unless otherwise notified.

Q.No	1	2	3	Total
Marks	25	25	25	75

Question 1

[25 Marks]

Evaluate the line integral $\oint_C (x^2 y \, dx + x y^2 \, dy)$ along the closed curve C enclosing region R . State and verify Green's Theorem $\oint_R ((\partial N / \partial x) - (\partial M / \partial y)) \, dA$.

Question 2

[25 Marks]

Evaluate the primary causes of the Indian Revolt of 1857. Analyze: (a) The Doctrine of Lapse, (b) Socio-religious anxieties, and (c) Immediate military triggers.

Question 3**[25 Marks]**

For an underdamped harmonic oscillator $m(d^2x / dt^2) + b(dx / dt) + kx = 0$, derive the general solution $x(t) = A e^{-\gamma t} \cos(\omega t + \phi)$ and find the damping ratio.
